25. What are the types of DNA polymerases present in *E.coli*? Write their functions.
26. Explain replication fork.
27. Explain bidirectional replication.
28. Explain continuous replication
29. Define okazaki fragments.
30. Why are RNA primers required
31. Describe what is meant by the antiparallel arrangement of DNA
32. Why is one strand of DNA synthesised discontinuously?
33. How is the faithfulness of DNA replication maintained? Write the name of the enzymes with its function.
34. Outline the experiment of Griffith.
35. Relate the experiments of Griffith and Avery.
36. Discuss Avery, Mc Cleod's experiment.
37. What was the motive of Avery and colleagues for conducting the experiment.
38. Differentiate between R and S strains of *Streptococcus pneumoniae*.
39. Describe the various characteristics of the Watson and Crick double helix model for DNA
40. Discuss the various bonds present in DNA double helix
41. Discuss various forms of DNA double helix structure.
42. Diagramatically explain the DNA double helix structure
43. Draw a four base pair segment of a DNA molecule, including each nucleotide and associated bonds involved in the maintenance of the double helix.
44. Diagrammatically represent the results of Meselson and Stahls experiment.
45. Explain how Meselson's and Stahl ruled out dispersive model of replication.
46. Tabulate the enzymes involved in DNA replication with their functions
47. Explain Elongation of DNA during replication

Student Activity

1. Fun with beads – students will understand the concept of polymer – polynucleotide and different sequences of DNA by preparing a chain of 20 beads of four different colours.
2. Prepare a model of DNA
3. Supercoiling of DNA – students will hold the ends of the rubber band and twist it. The two ends will be joined to feel the stress of coiling relieved due to supercoiling.
4. On paper replicate the following segment of DNA

   5′ATCGGCTACGTTCAC3′

   3′TAGCCGATGCAAGTG5′

   Show the direction of replication of the new strands and explain what the lagging and leading strands are? Explain how this is semiconservative replication. Are the new strands identical to the original segment DNA?

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